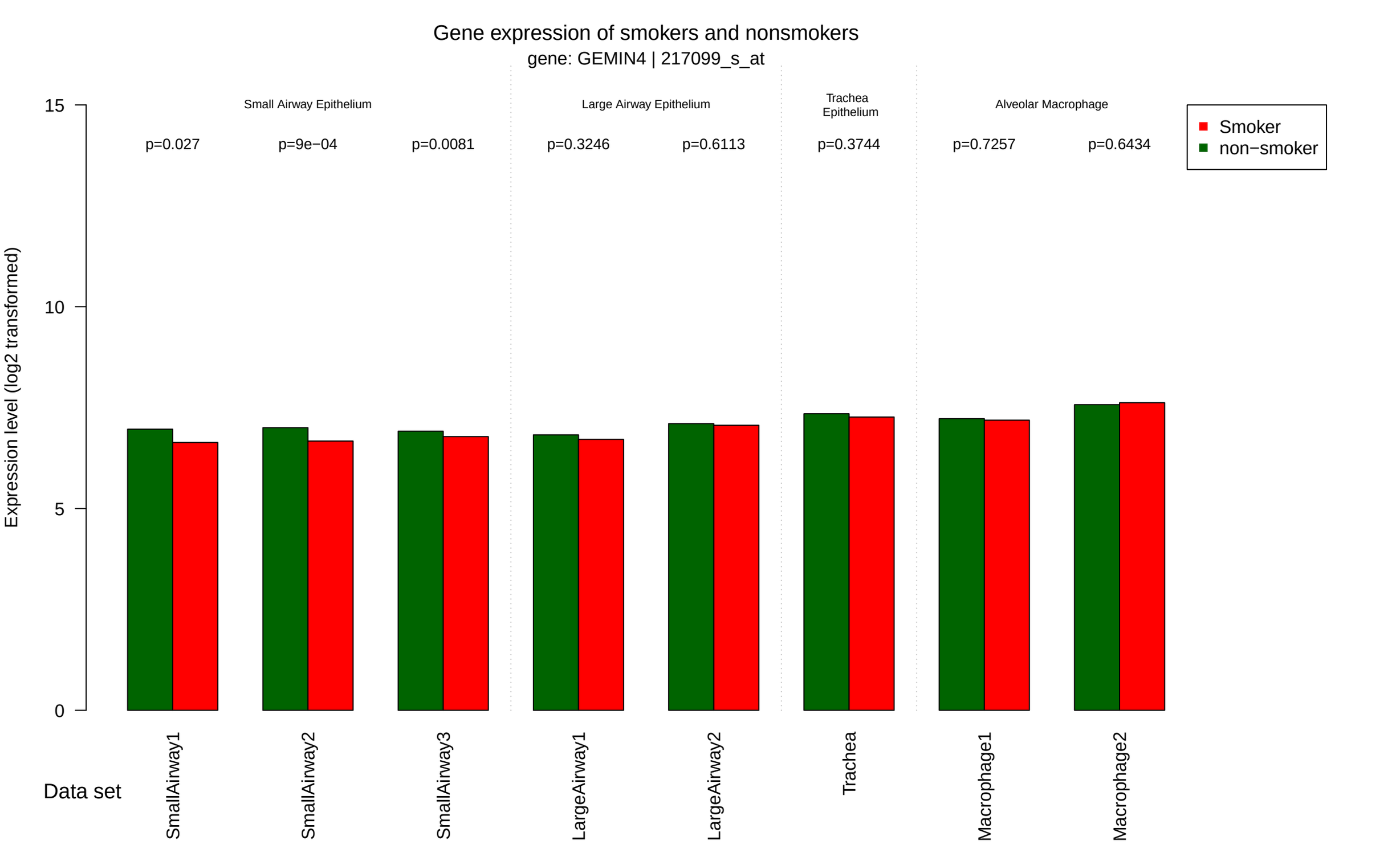
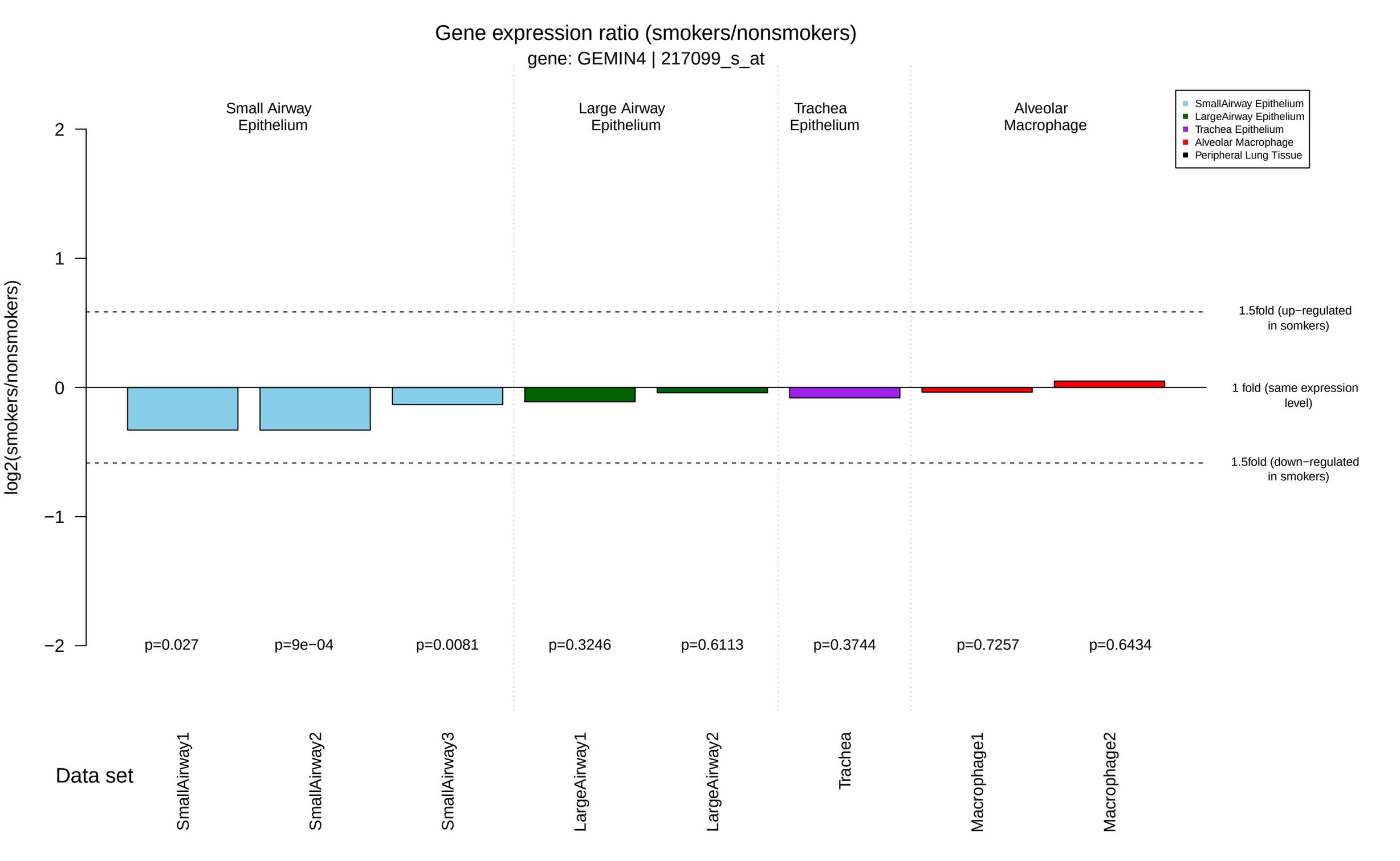
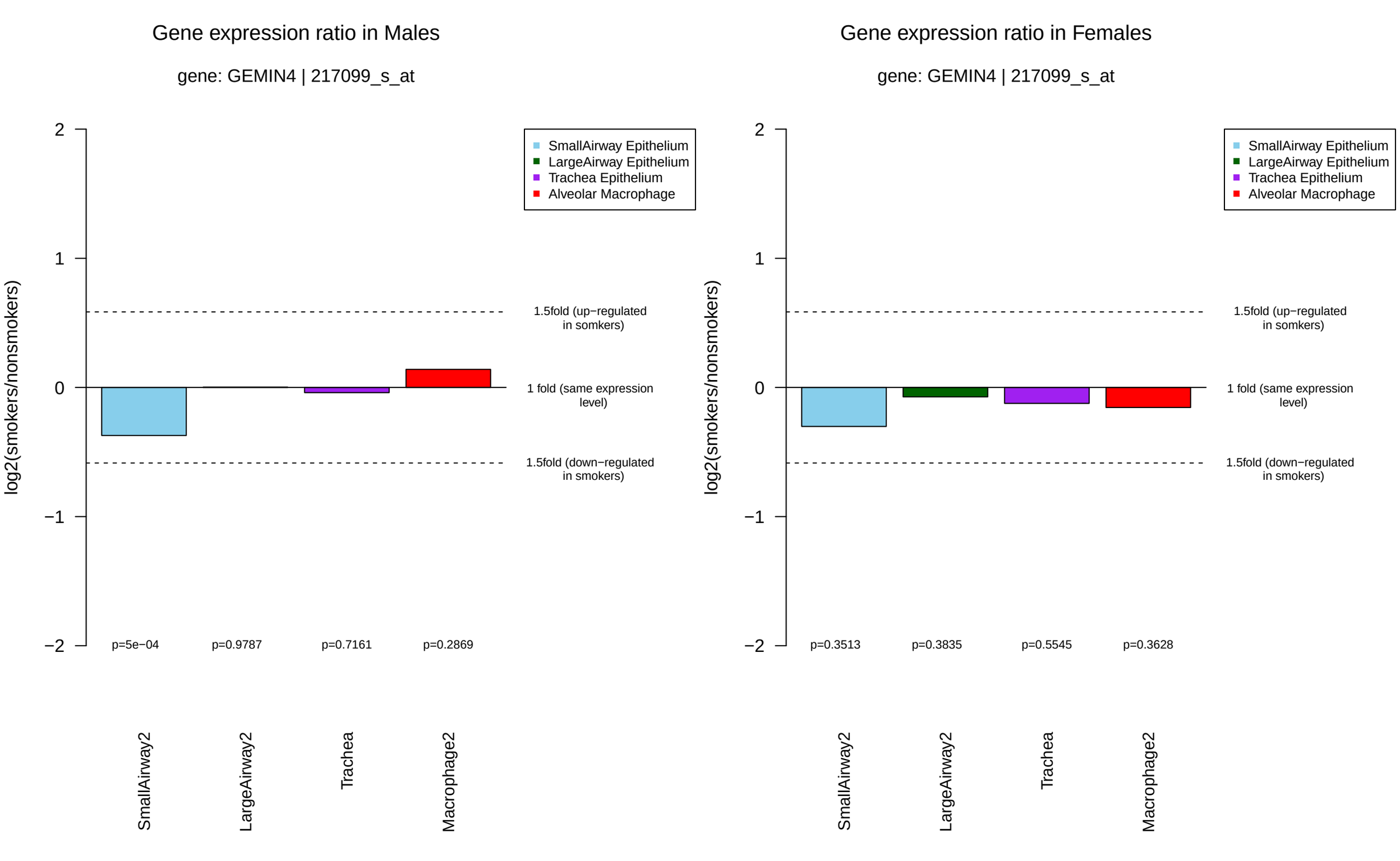
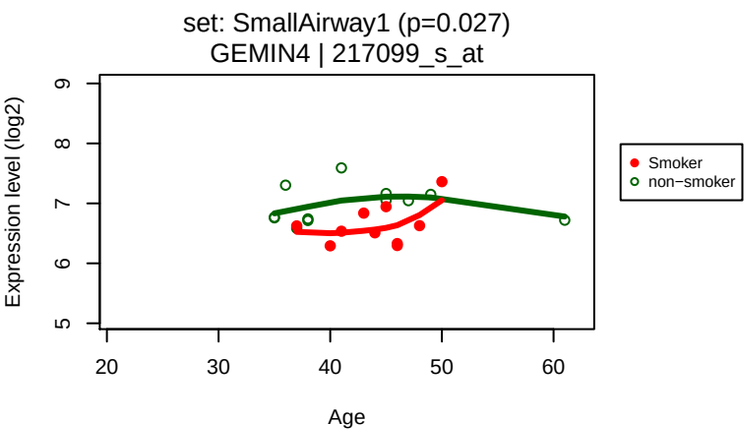
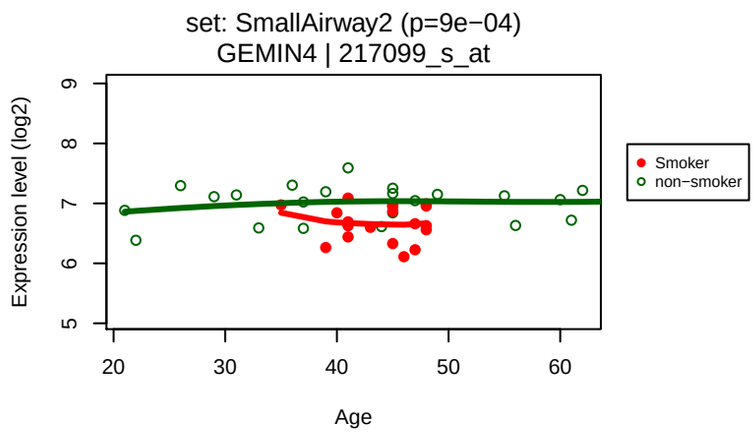
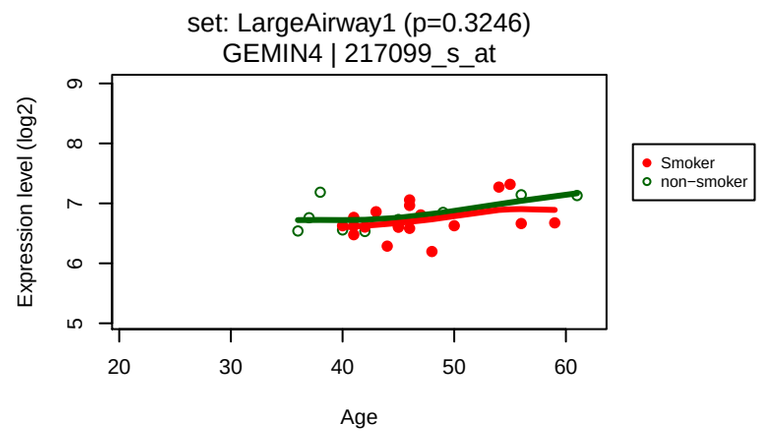

Gene expression of smokers and nonsmokers
gene: GEMIN4 | 217099_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.027
p=9e−04
p=0.0081
p=0.3246
p=0.6113
p=0.3744
p=0.7257
p=0.6434
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

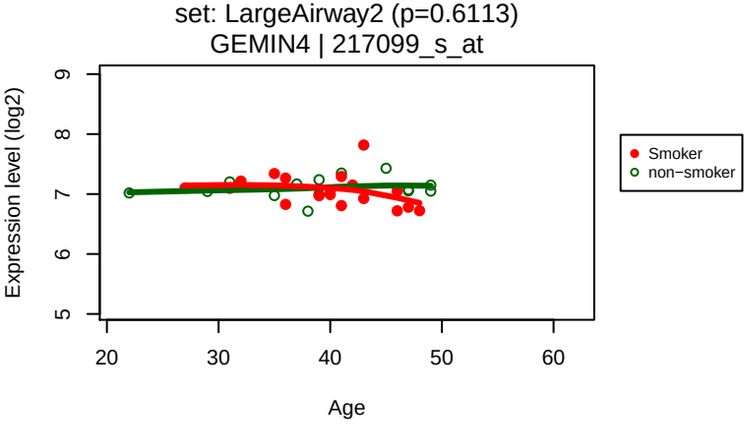

Gene expression ratio (smokers/nonsmokers)
gene: GEMIN4 | 217099_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.027
p=9e−04
p=0.0081
p=0.3246
p=0.6113
p=0.3744
p=0.7257
p=0.6434
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

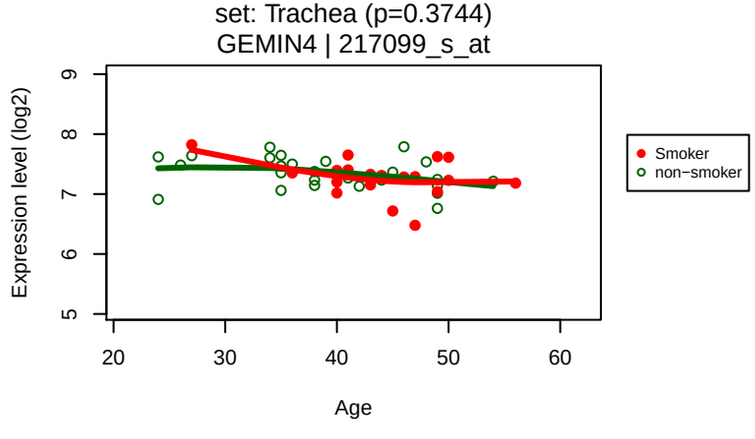

Gene expression ratio in Males
gene: GEMIN4 | 217099_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.9787
p=0.7161
p=0.2869
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GEMIN4 | 217099_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3513
p=0.3835
p=0.5545
p=0.3628
SmallAirway2
LargeAirway2
Trachea
Macrophage2

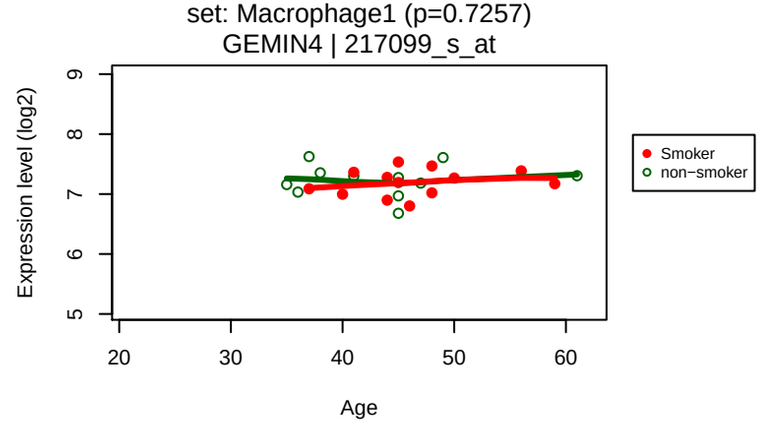

set: SmallAirway1 (p=0.027)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker

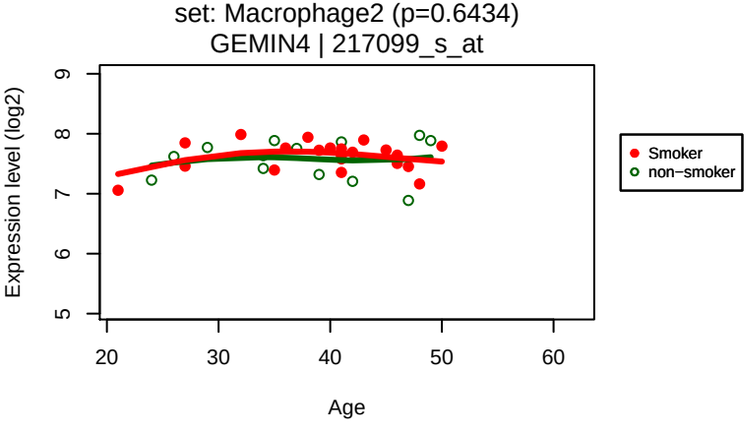

set: SmallAirway2 (p=9e−04)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker

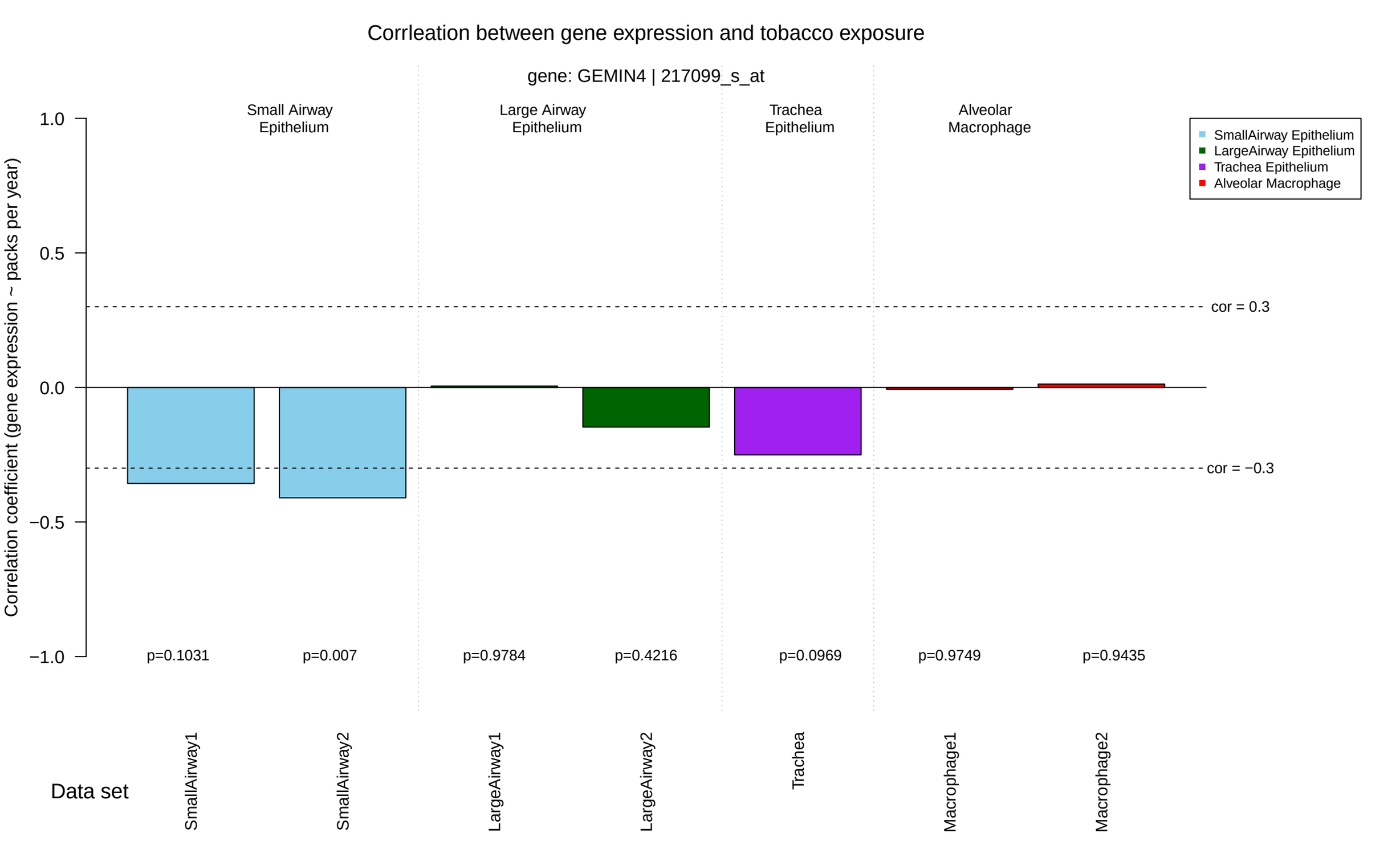

set: LargeAirway1 (p=0.3246)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6113)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3744)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7257)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6434)
GEMIN4 | 217099_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GEMIN4 | 217099_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1031
p=0.007
p=0.9784
p=0.4216
p=0.0969
p=0.9749
p=0.9435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2